lab02

1 - As estatísticas suficientes são o número total de voos por grupo e o número de voos atrasados (ARRIVAL_DELAY > 10) por grupo

2-

```r
getStats = function(input, pos) {
  input %>%
    # a.
    filter(AIRLINE %in% c("AA", "DL", "UA", "US")) %>%
    # b.
    drop_na(YEAR, MONTH, DAY, AIRLINE, ARRIVAL_DELAY) %>%
    # c.
    group_by(YEAR, MONTH, DAY, AIRLINE) %>%
    # d.
    summarise(
      voos_totais = n(),
      voos_atrasados = sum(ARRIVAL_DELAY > 10),
      .groups = "drop"
    )
}
```

3 -

```r
library(readr)
library(tidyr)
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
# a.
colunas_interesse = cols_only(
  YEAR = col_integer(),
  MONTH = col_integer(),
  DAY = col_integer(),
  AIRLINE = col_character(),
  ARRIVAL_DELAY = col_double()
)
# b.
f_callback = DataFrameCallback$new(getStats)
# c.
dados_parciais = read_csv_chunked(
  file = "flights.csv",
  callback = f_callback,
  chunk_size = 100000,
  col_types = colunas_interesse
)
```

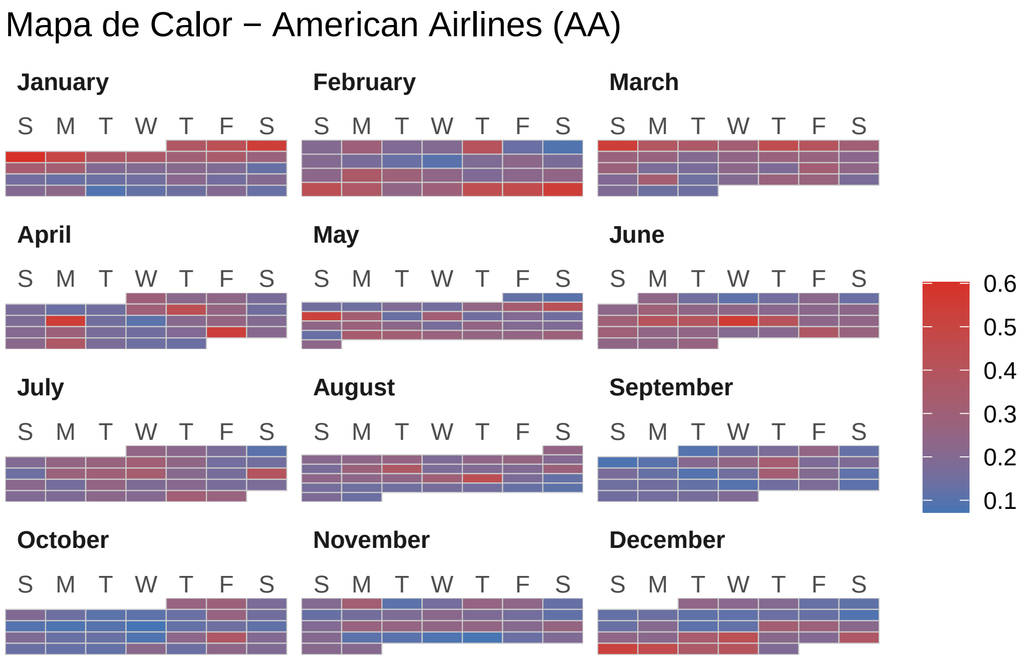
4 -

```r
computeStats = function(df_parcial) {
  df_parcial %>%
    group_by(YEAR, MONTH, DAY, AIRLINE) %>%
    summarise(
      voos_totais = sum(voos_totais),
      voos_atrasados = sum(voos_atrasados),
      .groups = "drop"
    ) %>%
    # a.
    mutate(
      Cia = AIRLINE,
      Data = as.Date(paste(YEAR, MONTH, DAY, sep = "-")),
      Perc = voos_atrasados / voos_totais
    ) %>%
```

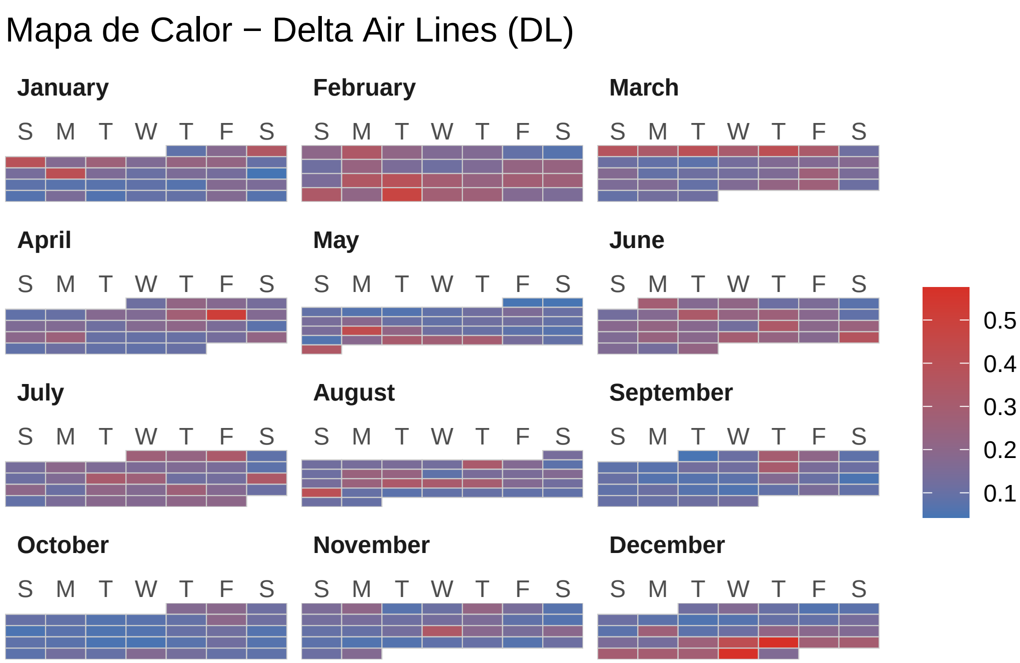
```r
    # b.
    select(Cia, Data, Perc)
}

stats_finais = computeStats(dados_parciais)
```

5 -

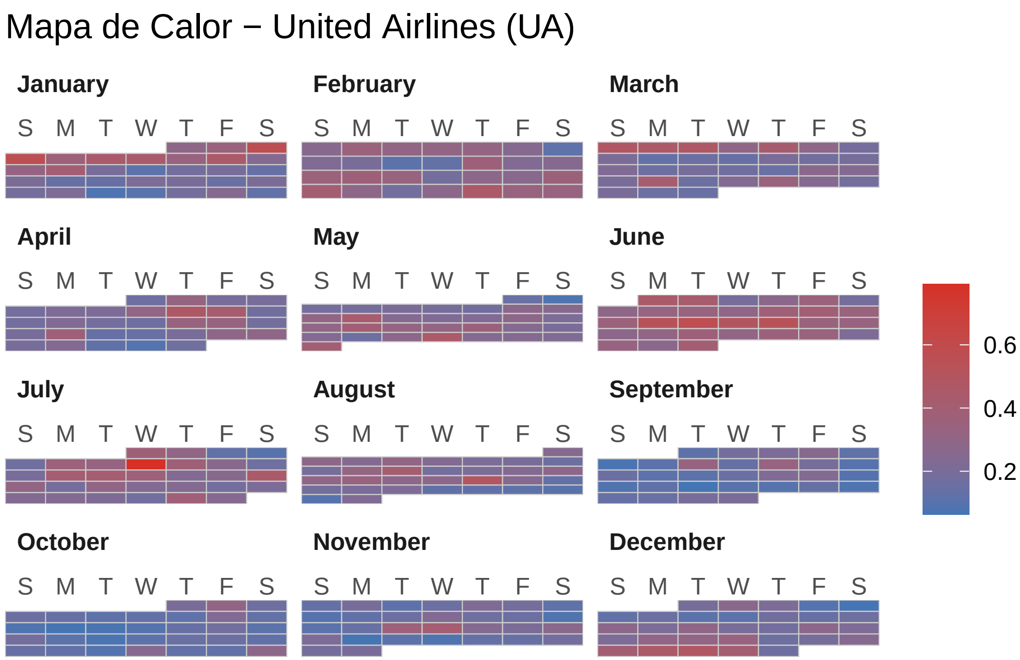
```r
# a.
library(ggplot2)
library(ggcal)
# b.
pal = scale_fill_gradient(low = "#4575b4", high = "#d73027", name = "% Atraso")
# c.
baseCalendario = function(stats, cia) {
  sub_stats = stats %>%
    filter(Cia == cia)

  ggcal(sub_stats$Data, sub_stats$Perc)
}
# d.
cAA = baseCalendario(stats_finais, "AA")
cDL = baseCalendario(stats_finais, "DL")
cUA = baseCalendario(stats_finais, "UA")
cUS = baseCalendario(stats_finais, "US")
# e.
cAA + pal + ggtitle("Mapa de Calor - American Airlines (AA)")
```



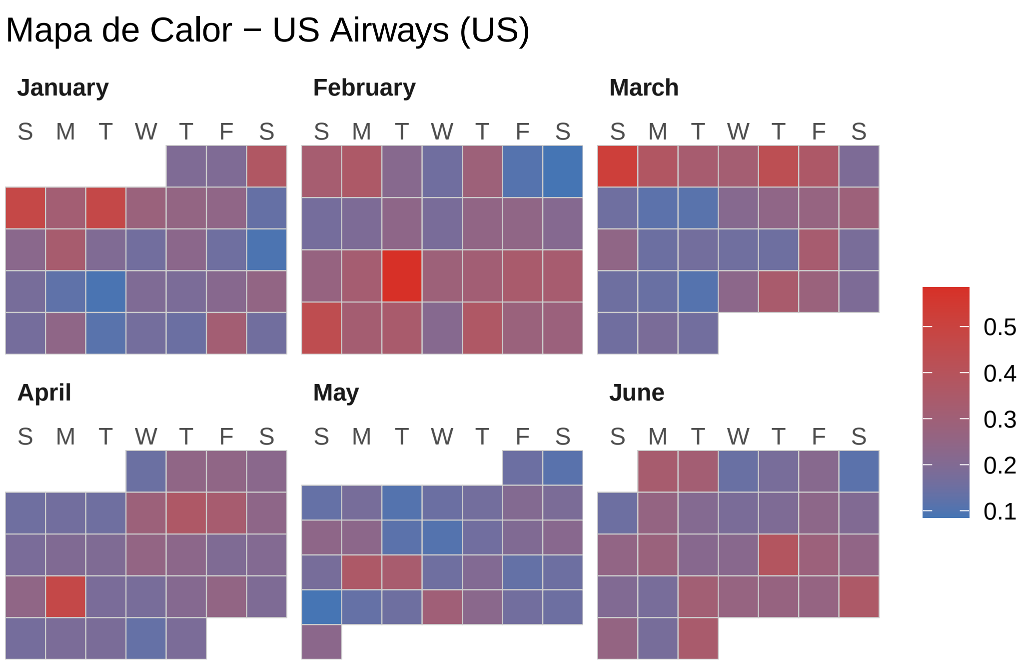

```r
cDL + pal + ggtitle("Mapa de Calor - Delta Air Lines (DL)")
```




```r
cUA + pal + ggtitle("Mapa de Calor - United Airlines (UA)")
```




```r
cUS + pal + ggtitle("Mapa de Calor - US Airways (US)")
```




```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-24 21:52:13 -03"
```